RV College of Engineering®

Academic Year 2022-2023 (Odd Semester)

USN:

Department of Artificial Intelligence and Machine Learning

rse Code:21AI52
: V

Date: 08.01.2024
Duration: 90 Minutes

CIE-I
Artificial Intelligence and Machine Learning

Answer all Questions

o	Questions	M	BT	CO
	For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate. a. An agent that senses only partial information about the state cannot be perfectly rational. b. There exists a task environment in which every agent is rational. c. The input to an agent program is the same as the input to the agent function. d. Suppose an agent selects its action uniformly at random from the set of possible actions. There exists a deterministic task environment in which this agent is rational. Ans:- yes e. Every agent is rational in an unobservable environment. Ans:- no f. A perfectly rational poker-playing agent never loses. Ans:- no.	06	L3	CO1
b)	Define Rational Agent (RA) and discuss the usage of RA in the following applications: a. Healthcare b. Transportation	04	L2	CO1
a)	For each of the following agents, develop a PEAS description of the task environment: a. Robot soccer player b. Internet book-shopping agent c. Autonomous Mars rover d. Mathematician's theorem-proving assistant.	06	L3	CO2
b)	Justify with reasons how utility-based agents is different from model-based reflex agents with appropriate diagrams.	04	L2	CO1
	Given a dataset with the following features (Age, Income, and Education) and a target variable (Buy: Yes/No). Write a pseudo code and Illustrate the Decision Tree algorithm to build a model. Should write tree also.	10	L4	CO3

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	For the given data, calculate formula is right but sign mistake (a) Entropy (b) Information Gain for the Outlook Feature	10	L3	CO5
a)	Imagine a binary classification problem where a Decision Tree model achieves 90% accuracy on the training set but only 75% accuracy on the test set. Analyze approaches to estimate the generalization performance of a model that can be used for model selection. Methods of generalization	05	L3	CO2
b)	Explain the concept of k-fold cross-validation? Provide an example to illustrate its application in Machine Learning?	05	L3	CO5

Day	Outlook	Temp	Humidity	Wind	Play Volleyball
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Rain	Mild	High	Strong	No
D13	Overcast	Hot	Normal	Weak	Yes
D14	Overcast	Mild	High	Strong	Yes

urse Outcome	
01	Explain and apply AI and ML algorithms to address various requirements of real-world problems.
02	Design and develop AI and ML solutions to benefit society, science, and industry.
03	Use modern tools to create AI and ML solutions.
04	Demonstrate effective communication through team presentations and reports to analyze the impact of AI and ML solutions on society and nature.
05	Conduct performance evaluation, modeling, and validation of AI and ML solutions benefiting lifelong learning.

M-Marks, BT-Blooms Taxonomy Levels, CO-Course Outcomes

Marks stribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	14	11	10	--	15		08	32	10	--	--